LAB: 3

Title: To implement and visualize Midpoint Circle Algorithm using Turbo C++

Program:

```
#include <graphics.h>
#include <conio.h>
#include <stdio.h>

int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "C:\\TC\\BGI");

    int xc, yc, r;
    printf("Enter the center of the circle (xc yc): ");
    scanf("%d %d", &xc, &yc);
    printf("Enter the radius of the circle: ");
    scanf("%d", &r);

    int x = 0;
    int y = r;
    int p = 1 - r;

    do {
        putpixel(xc + x, yc + y, WHITE);
        putpixel(xc - x, yc + y, WHITE);
        putpixel(xc + x, yc - y, WHITE);
        putpixel(xc - x, yc - y, WHITE);
        putpixel(xc + y, yc + x, WHITE);
        putpixel(xc - y, yc + x, WHITE);
        putpixel(xc + y, yc - x, WHITE);
        putpixel(xc - y, yc - x, WHITE);

        x++;
```

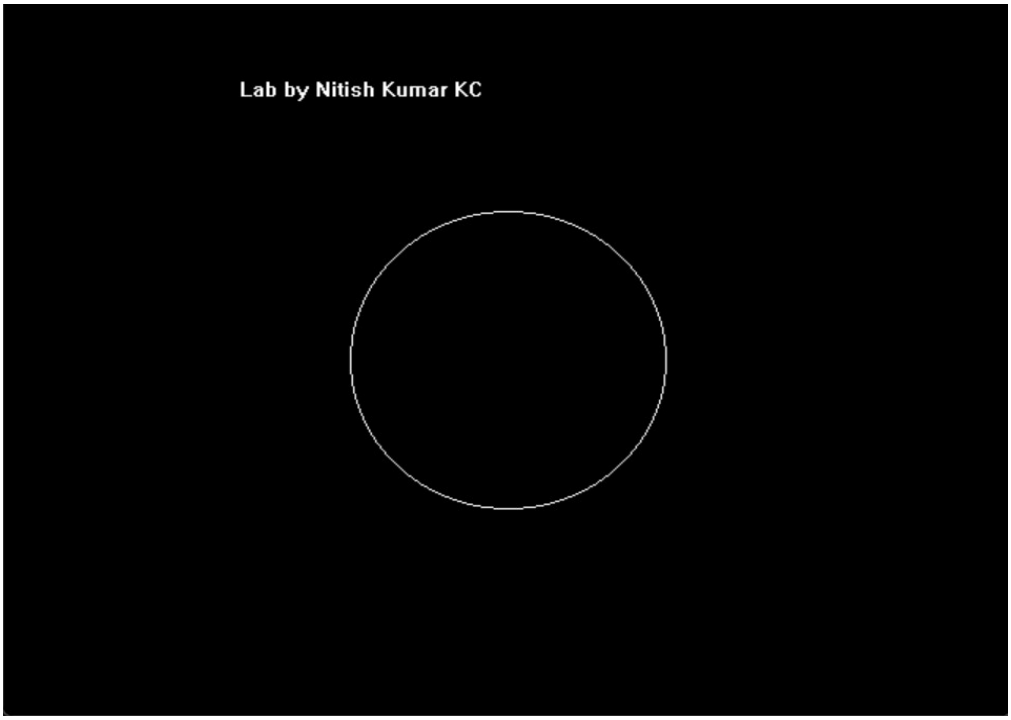
```
        if (p < 0)
            p = p + 2 * x + 1;
        else {
            y--;
            p = p + 2 * (x - y) + 1;
        }
    } while (x <= y);

    outtextxy(150, 50, "Lab by Nitish Kumar KC");

    getch();
    closegraph();
    return 0;
}
```

Output:

```
Enter the center of the circle (xc yc): 320
240
Enter the radius of the circle: 100
```



LAB: 4

Title: To implement and visualize Midpoint Ellipse Algorithm using Turbo C++

Program:

```
#include <graphics.h>
#include <conio.h>
#include <stdio.h>
int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, "C:\\TC\\BGI");

    int xc, yc, a, b;
    printf("Enter the center of the ellipse (xc yc): ");
    scanf("%d %d", &xc, &yc);
    printf("Enter the minor axis (a) and major axis (b): ");
    scanf("%d %d", &a, &b);

    int x = 0;
    int y = b;
    int a2 = a * a;
    int b2 = b * b;
    int two_a2 = 2 * a2;
    int two_b2 = 2 * b2;
    int px = 0;
    int py = two_a2 * y;
    int p1 = b2 - (a2 * b) + (a2 / 4);
    // Region 1
    do {
        putpixel(xc + x, yc + y, WHITE);
        putpixel(xc - x, yc + y, WHITE);
        putpixel(xc + x, yc - y, WHITE);
        putpixel(xc - x, yc - y, WHITE);
```

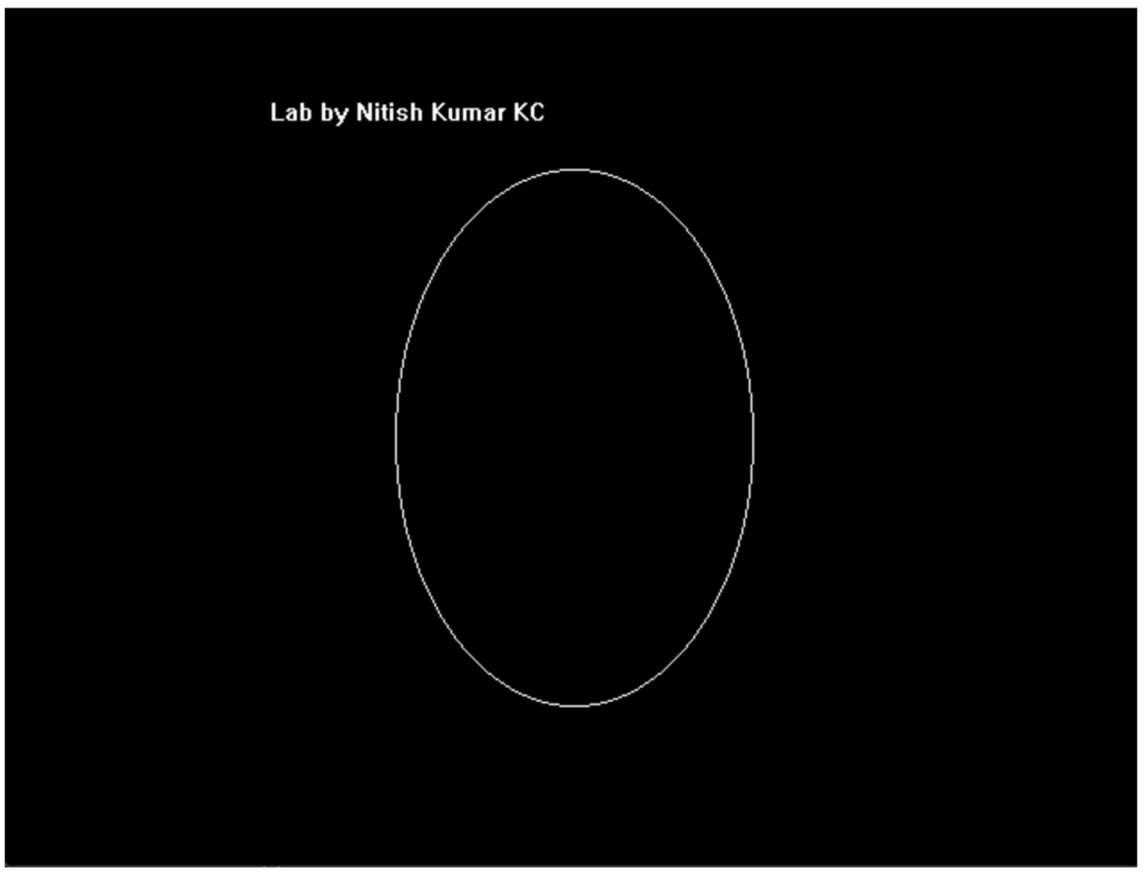
```
  x++;
    px += two_b2;
    if (p1 < 0)
        p1 += b2 + px;
    else {
        y--;
        py -= two_a2;
        p1 += b2 + px - py;
    }
} while (px < py);

// Region 2
int p2 = b2 * (x + 0.5) * (x + 0.5) + a2 * (y - 1) * (y - 1) - a2 * b2;
do {
    putpixel(xc + x, yc + y, WHITE);
    putpixel(xc - x, yc + y, WHITE);
    putpixel(xc + x, yc - y, WHITE);
    putpixel(xc - x, yc - y, WHITE);

    y--;
    py -= two_a2;
    if (p2 > 0)
        p2 += a2 - py;
    else {
        x++;
        px += two_b2;
        p2 += a2 - py + px;
    }
} while (y >= 0);

outtextxy(150, 50, "Lab by Nitish Kumar KC");
```

```
    getch();
    closegraph();
    return 0;
}
```

Output:

```
Enter the center of the ellipse (xc yc): 320
240
Enter the minor axis (a) and major axis (b): 100
150
```